Applied Machine Learning

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- ML models have predictive / estimation capabilities.

- To develop these capability in ML models, we need to train them using good amount of data and an appropriate algorithm

eg.

ML model	data	algorithm
- house price prediction	housing data	regression
- face recognition	images of face	classification
- customer segmentation	customer data	clustering

Human Learning → 1. observe information from data
2. analyse, understand patterns
3. get insights
4. remember the learnings
5. use these learnings to perform tasks

correctly performed task → learning was correct → remember the learning deeply

incorrectly performed → learning is incorrectly → correct the learning by redoing previous steps

Machine learning → 1. observe info from data
2. analyse, understand patterns
3. get insights, learn the task
4. remember the learnings
5. perform the task based on learnings

correctly performed
↓
learning was correct
↓
remember this learning

incorrectly performed
↓
learning is incorrect
↓
correct mistakes through better learning / correct learnings

Human learning and machine learning, both are <u>cyclic process.</u>

To create any ML system we need

1) data : relevant for given use case
2) algorithm : relevant for given use case
3) computer : for performing math computations

* Choosing correct data and correct algorithm for given ML use case is the main skill of AI engineer / data scientist.

Data - it has information required to perform the given task

Algorithm - it tries to understand the information available in the data
tries to learn the given task
learns to convert input into output

AI/ML solution	Data
- house price prediction	area, rooms, location, price
- chatbot	que and ans text
- face recognition	face images, label - yes/no
- cancer cell detection	scan images, label - yes/no

- We train our ML model with chosen data and algorithm.
- we check its performance on the task after training
- if performance is satisfactory, we start using model on completely new data/examples.

- There are multiple algorithms available for ML model training
- Each algorithm has a unique way of training
- The choice of algorithm mainly depends on 2 things
1) task we want to automate 2) type of available data

At first we choose one of the below type of algorithm

There are two major types of techniques / types of ML algorithms

1) Supervised ML

2) Unsupervised ML

In ML we have data for model training from which the model learns the given task
- this data is called <u>training data</u>

eg.
house price prediction

	area	rooms	price
P_1	1100	2	70
P_2	1200	2	72
P_3	1000	1	65
P_4	850	1	61
⋮	⋮	⋮	⋮
P_n	⋮	⋮	⋮

← train data example

} total 'n' datapoints

$$P = \underbrace{\overset{\text{area}}{1200} \quad \overset{\text{rooms}}{2}}_{\text{input features}} \quad \underbrace{\overset{\text{price}}{72}}_{\text{targed output}}$$

area
rooms → [ML model] → price

✓ ✓

For given task when dataset has features and target both available, then its called <u>Labelled</u> dataset.

When labelled data is to be used for model training, we choose <u>supervised ML algorithms</u>.